

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented REST APIs using **Java Spring Boot** for cross-service data exchange, incorporating request validation and circuit-breaker patterns.
- Managed production support for enterprise applications, addressing P1/P2 incidents and performing root cause analysis using **Splunk** and **Dynatrace** to maintain service level agreements.
- Developed Python scripts to automate data ingestion and transformation tasks for scheduled reporting, improving data pipeline efficiency by **18%**.
- Contributed to the preparation of design reports and technical specifications for water engineering projects, including drawings and calculations.
- Gained exposure to building clean water network models using various modelling software, supporting feasibility studies and detailed design phases.
- Assisted senior engineers in developing detailed project plans, focusing on budgets, schedules, and resource allocation for infrastructure projects.
- Containerized Java Spring Boot applications using **Docker** and deployed them to **AWS EC2** instances, configuring basic resource limits for predictable rollouts.

SKILLS

Engineering Principles – Water Engineering, Civil Engineering, Environmental Engineering, Infrastructure Projects, Feasibility Studies, Design Reports, Technical Specifications, Contract Documents, Health & Safety Legislation, Quality Management Systems.

Software & Tools – AutoCAD, Civil 3D, GIS (ArcGIS, QGIS), Hydraulic Modelling Software (Epanet, Synergi, MicroDrainage, HEC-RAS), **Python**, **Java**, **Spring Boot**, **SQL**, **AWS** (EC2, S3, Lambda, CloudWatch), **Docker**, **Linux**, **Git**, **Jenkins**, **Jira**, **Confluence**, **Splunk**, **Dynatrace**, **Power BI**, **Tableau**, R / RStudio, pandas, NumPy, scikit-learn, Flask, FastAPI.

Data Analysis & BI – Data Modelling, Data Warehousing, ETL, DAX Measures, Power Query, Statistical Analysis, Exploratory Data Analysis, A/B Testing, Time-Series Analysis, Business Intelligence.

Project Management – Project Planning, Budget Management, Schedule Management, Resource Allocation, Risk Mitigation, Stakeholder Communication, Client Collaboration, Team Collaboration.

Databases – **PostgreSQL**, MySQL, Oracle.

PROJECTS

Water Network Analysis and Design Tool

- Developed a Python-based tool to assist in the analysis of clean water network models, integrating with **Epanet** for simulation outputs.
- Designed and implemented data validation scripts using **pandas** to ensure input data quality for hydraulic modelling, reducing data entry errors by approximately **15%**.
- Created interactive dashboards using **Power BI** to visualize network performance metrics and simulation results for project stakeholders.

Java Application for Infrastructure Project Documentation

- Built a **Java Spring Boot** application to manage and track documentation for infrastructure projects, including design reports and statutory approval submissions.
- Implemented RESTful APIs for system integration and data retrieval, ensuring seamless data flow between project management modules and a **PostgreSQL** database.
- Containerized the application using **Docker** for consistent deployment across development and testing environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore,
India**

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)